Graph encoders: what each mechanism adds

MPNN

local messages

$$m_i = \sum_{j \in \mathcal{N}(i)} M_{ij}$$

captures

functional groups
short-range chemistry

main caveat

long-range context
is indirect

GPS

local + global

message passing
+ self-attention

captures

non-local context
for scaffold transfer

main caveat

cost and overfitting
risk increase

TIMP

typed channels

$$m_{ij} = m_{ij}^{disp} + m_{ij}^{polar}$$

captures

dispersive effects
polar / H-bond effects

main caveat

depends on charge
and edge features

All three encoders return molecule vectors; they differ only in how atom-level information is mixed.